SG No. 201 T_{h}^2 $Pn\bar{3}$ [cubic]

* plus set: $+[0,\ 0,\ 0]$
* Wyckoff site: `2a`, site symmetry: `23.`

Table 1: Wyckoff bond: `6a@2a`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-2,-3,4]`
2	$[0,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,-6,-7,8]`
3	$[0,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[9,-10,-11,12]`
4	$[-X,\ 0,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,-14,-15,16]`
5	$[0,\ -X,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[17,-18,-19,20]`
6	$[0,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[21,-22,-23,24]`

Table 2: Wyckoff bond: `8b@2a`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,5,9]`
2	$[-X,\ -X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,7,12]`
3	$[-X,\ X,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,8,10]`
4	$[X,\ -X,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[4,6,11]`
5	$[-X,\ -X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,17,21]`
6	$[X,\ X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[14,19,24]`
7	$[X,\ -X,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[15,20,22]`
8	$[-X,\ X,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[16,18,23]`

Table 3: Wyckoff bond: `12c@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-2]`
2	$[-X,\ Y,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,-4]`
3	$[0,\ X,\ Y]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,-6]`
4	$[0,\ -X,\ Y]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[7,-8]`
5	$[Y,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[9,-10]`
6	$[Y,\ 0,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[11,-12]`
7	$[-X,\ -Y,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,-14]`
8	$[X,\ -Y,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[15,-16]`
9	$[0,\ -X,\ -Y]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[17,-18]`
10	$[0,\ X,\ -Y]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[19,-20]`
11	$[-Y,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[21,-22]`
12	$[-Y,\ 0,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[23,-24]`

Table 4: Wyckoff bond: `24d@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1]`

continued ...

Table 4

No.	vector	center	mapping
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[4]`
5	$[Z, X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[8]`
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[12]`
13	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[13]`
14	$[X, Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[14]`
15	$[X, -Y, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[15]`
16	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[16]`
17	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[17]`
18	$[-Z, X, Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[18]`
19	$[Z, X, -Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[19]`
20	$[Z, -X, Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[20]`
21	$[-Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[21]`
22	$[Y, -Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[22]`
23	$[-Y, Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[23]`
24	$[Y, Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[24]`

* Wyckoff site: `4b`, site symmetry: `.-3.`

Table 5: Wyckoff bond: `4a@4b`, bond symmetry: `.-3.`

No.	vector	center	mapping
1	$[X, X, X]$	$[0, 0, 0]$	`[1,5,9,-13,-17,-21]`
2	$[-X, -X, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,7,12,-14,-19,-24]`
3	$[-X, X, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,8,10,-15,-20,-22]`
4	$[X, -X, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,6,11,-16,-18,-23]`

Table 6: Wyckoff bond: `12b@4b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-13]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-14]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,-15]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-16]`
5	$[Z, X, Y]$	$[0, 0, 0]$	`[5,-17]`
6	$[Z, -X, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-18]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[7,-19]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-20]`
9	$[Y, Z, X]$	$[0, 0, 0]$	`[9,-21]`

continued ...

Table 6

No.	vector	center	mapping
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[10,-22]`
11	$[Y, -Z, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,-23]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[12,-24]`

* Wyckoff site: `4c`, site symmetry: `.-3.`

Table 7: Wyckoff bond: `4a@4c`, bond symmetry: `.-3.`

No.	vector	center	mapping
1	$[X, X, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,5,9,-13,-17,-21]`
2	$[-X, -X, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,7,12,-14,-19,-24]`
3	$[-X, X, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,8,10,-15,-20,-22]`
4	$[X, -X, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,6,11,-16,-18,-23]`

Table 8: Wyckoff bond: `12b@4c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-13]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-14]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-15]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-16]`
5	$[Z, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-17]`
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[6,-18]`
7	$[-Z, -X, Y]$	$\left[0, 0, \frac{1}{2}\right]$	`[7,-19]`
8	$[-Z, X, -Y]$	$\left[0, \frac{1}{2}, 0\right]$	`[8,-20]`
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,-21]`
10	$[-Y, Z, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[10,-22]`
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[11,-23]`
12	$[-Y, -Z, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[12,-24]`

* Wyckoff site: `6d`, site symmetry: `222..`

Table 9: Wyckoff bond: `6a@6d`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,-2,-3,4]`
2	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,-6,-7,8]`
3	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[9,-10,-11,12]`
4	$[-X, 0, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[13,-14,-15,16]`
5	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[17,-18,-19,20]`
6	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[21,-22,-23,24]`

Table 10: Wyckoff bond: `6b@6d`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[0,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,2,-3,-4]`
2	$[X,\ 0,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[5,6,-7,-8]`
3	$[0,\ X,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9,10,-11,-12]`
4	$[0,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,14,-15,-16]`
5	$[-X,\ 0,\ 0]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[17,18,-19,-20]`
6	$[0,\ -X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[21,22,-23,-24]`

Table 11: Wyckoff bond: `6c@6d`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[0,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,-2,3,-4]`
2	$[0,\ 0,\ X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[5,-6,7,-8]`
3	$[X,\ 0,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9,-10,11,-12]`
4	$[0,\ -X,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,-14,15,-16]`
5	$[0,\ 0,\ -X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[17,-18,19,-20]`
6	$[-X,\ 0,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[21,-22,23,-24]`

Table 12: Wyckoff bond: `12d@6d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,-2]`
2	$[-X,\ Y,\ 0]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[3,-4]`
3	$[0,\ X,\ Y]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[5,-6]`
4	$[0,\ -X,\ Y]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[7,-8]`
5	$[Y,\ 0,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9,-10]`
6	$[Y,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[11,-12]`
7	$[-X,\ -Y,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,-14]`
8	$[X,\ -Y,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[15,-16]`
9	$[0,\ -X,\ -Y]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[17,-18]`
10	$[0,\ X,\ -Y]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[19,-20]`
11	$[-Y,\ 0,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[21,-22]`
12	$[-Y,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[23,-24]`

Table 13: Wyckoff bond: `12e@6d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[Y,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,-3]`
2	$[-Y,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[2,-4]`
3	$[X,\ Y,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[5,-7]`
4	$[X,\ -Y,\ 0]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[6,-8]`
5	$[0,\ X,\ Y]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9,-11]`
6	$[0,\ X,\ -Y]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[10,-12]`
7	$[-Y,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,-15]`

continued ...

Table 13

No.	vector	center	mapping
8	$[Y, 0, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[14,-16]`
9	$[-X, -Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[17,-19]`
10	$[-X, Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[18,-20]`
11	$[0, -X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[21,-23]`
12	$[0, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[22,-24]`

Table 14: Wyckoff bond: `12f@6d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[1,-4]`
2	$[0, -X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[2,-3]`
3	$[Y, 0, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[5,-8]`
4	$[Y, 0, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[6,-7]`
5	$[X, Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[9,-12]`
6	$[-X, Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[10,-11]`
7	$[0, -X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[13,-16]`
8	$[0, X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[14,-15]`
9	$[-Y, 0, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[17,-20]`
10	$[-Y, 0, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[18,-19]`
11	$[-X, -Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[21,-24]`
12	$[X, -Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[22,-23]`

Table 15: Wyckoff bond: `24g@6d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[12]`
13	$[-X, -Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[13]`
14	$[X, Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[14]`
15	$[X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[15]`
16	$[-X, Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[16]`
17	$[-Z, -X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[17]`
18	$[-Z, X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[18]`
19	$[Z, X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[19]`

continued ...

Table 15

No.	vector	center	mapping
20	$[Z, -X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[20]
21	$[-Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[21]
22	$[Y, -Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[22]
23	$[-Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[23]
24	$[Y, Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[24]

* Wyckoff site: 8e, site symmetry: .3.

Table 16: Wyckoff bond: 8a@8e, bond symmetry: .3.

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	[1,5,9]
2	$[-X, -X, X]$	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x\right]$	[2,7,12]
3	$[-X, X, -X]$	$\left[\frac{1}{2} - x, x, \frac{1}{2} - x\right]$	[3,8,10]
4	$[X, -X, -X]$	$\left[x, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[4,6,11]
5	$[-X, -X, -X]$	$[-x, -x, -x]$	[13,17,21]
6	$[X, X, -X]$	$\left[x + \frac{1}{2}, x + \frac{1}{2}, -x\right]$	[14,19,24]
7	$[X, -X, X]$	$\left[x + \frac{1}{2}, -x, x + \frac{1}{2}\right]$	[15,20,22]
8	$[-X, X, X]$	$\left[-x, x + \frac{1}{2}, x + \frac{1}{2}\right]$	[16,18,23]

Table 17: Wyckoff bond: 24b@8e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, x, \frac{1}{2} - x\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[4]
5	$[Z, X, Y]$	$[x, x, x]$	[5]
6	$[Z, -X, -Y]$	$\left[x, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{2} - x, x, \frac{1}{2} - x\right]$	[8]
9	$[Y, Z, X]$	$[x, x, x]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2} - x, x, \frac{1}{2} - x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[x, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x\right]$	[12]
13	$[-X, -Y, -Z]$	$[-x, -x, -x]$	[13]
14	$[X, Y, -Z]$	$\left[x + \frac{1}{2}, x + \frac{1}{2}, -x\right]$	[14]
15	$[X, -Y, Z]$	$\left[x + \frac{1}{2}, -x, x + \frac{1}{2}\right]$	[15]
16	$[-X, Y, Z]$	$\left[-x, x + \frac{1}{2}, x + \frac{1}{2}\right]$	[16]
17	$[-Z, -X, -Y]$	$[-x, -x, -x]$	[17]
18	$[-Z, X, Y]$	$\left[-x, x + \frac{1}{2}, x + \frac{1}{2}\right]$	[18]
19	$[Z, X, -Y]$	$\left[x + \frac{1}{2}, x + \frac{1}{2}, -x\right]$	[19]
20	$[Z, -X, Y]$	$\left[x + \frac{1}{2}, -x, x + \frac{1}{2}\right]$	[20]
21	$[-Y, -Z, -X]$	$[-x, -x, -x]$	[21]
22	$[Y, -Z, X]$	$\left[x + \frac{1}{2}, -x, x + \frac{1}{2}\right]$	[22]
23	$[-Y, Z, X]$	$\left[-x, x + \frac{1}{2}, x + \frac{1}{2}\right]$	[23]

continued ...

Table 17

No.	vector	center	mapping
24	$[Y,\, Z,\, -X]$	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, -x\right]$	`[24]`

* Wyckoff site: `12f`, site symmetry: `2..`

Table 18: Wyckoff bond: `12a@12f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\, X,\, Y]$	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[1,-4]`
2	$[0,\, -X,\, Y]$	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[2,-3]`
3	$[Y,\, 0,\, X]$	$\left[\frac{1}{4},\, x,\, \frac{1}{4}\right]$	`[5,-8]`
4	$[Y,\, 0,\, -X]$	$\left[\frac{1}{4},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	`[6,-7]`
5	$[X,\, Y,\, 0]$	$\left[\frac{1}{4},\, \frac{1}{4},\, x\right]$	`[9,-12]`
6	$[-X,\, Y,\, 0]$	$\left[\frac{1}{4},\, \frac{1}{4},\, \frac{1}{2}-x\right]$	`[10,-11]`
7	$[0,\, -X,\, -Y]$	$\left[-x,\, \frac{3}{4},\, \frac{3}{4}\right]$	`[13,-16]`
8	$[0,\, X,\, -Y]$	$\left[x+\frac{1}{2},\, \frac{3}{4},\, \frac{3}{4}\right]$	`[14,-15]`
9	$[-Y,\, 0,\, -X]$	$\left[\frac{3}{4},\, -x,\, \frac{3}{4}\right]$	`[17,-20]`
10	$[-Y,\, 0,\, X]$	$\left[\frac{3}{4},\, x+\frac{1}{2},\, \frac{3}{4}\right]$	`[18,-19]`
11	$[-X,\, -Y,\, 0]$	$\left[\frac{3}{4},\, \frac{3}{4},\, -x\right]$	`[21,-24]`
12	$[X,\, -Y,\, 0]$	$\left[\frac{3}{4},\, \frac{3}{4},\, x+\frac{1}{2}\right]$	`[22,-23]`

Table 19: Wyckoff bond: `12b@12f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\, 0,\, 0]$	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[1,4]`
2	$[-X,\, 0,\, 0]$	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[2,3]`
3	$[0,\, X,\, 0]$	$\left[\frac{1}{4},\, x,\, \frac{1}{4}\right]$	`[5,8]`
4	$[0,\, -X,\, 0]$	$\left[\frac{1}{4},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	`[6,7]`
5	$[0,\, 0,\, X]$	$\left[\frac{1}{4},\, \frac{1}{4},\, x\right]$	`[9,12]`
6	$[0,\, 0,\, -X]$	$\left[\frac{1}{4},\, \frac{1}{4},\, \frac{1}{2}-x\right]$	`[10,11]`
7	$[-X,\, 0,\, 0]$	$\left[-x,\, \frac{3}{4},\, \frac{3}{4}\right]$	`[13,16]`
8	$[X,\, 0,\, 0]$	$\left[x+\frac{1}{2},\, \frac{3}{4},\, \frac{3}{4}\right]$	`[14,15]`
9	$[0,\, -X,\, 0]$	$\left[\frac{3}{4},\, -x,\, \frac{3}{4}\right]$	`[17,20]`
10	$[0,\, X,\, 0]$	$\left[\frac{3}{4},\, x+\frac{1}{2},\, \frac{3}{4}\right]$	`[18,19]`
11	$[0,\, 0,\, -X]$	$\left[\frac{3}{4},\, \frac{3}{4},\, -x\right]$	`[21,24]`
12	$[0,\, 0,\, X]$	$\left[\frac{3}{4},\, \frac{3}{4},\, x+\frac{1}{2}\right]$	`[22,23]`

Table 20: Wyckoff bond: `24c@12f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[1]`
2	$[-X,\, -Y,\, Z]$	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[2]`
3	$[-X,\, Y,\, -Z]$	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[3]`
4	$[X,\, -Y,\, -Z]$	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[4]`
5	$[Z,\, X,\, Y]$	$\left[\frac{1}{4},\, x,\, \frac{1}{4}\right]$	`[5]`

continued ...

Table 20

No.	vector	center	mapping
6	$[Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[12]
13	$[-X, -Y, -Z]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[13]
14	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[14]
15	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[15]
16	$[-X, Y, Z]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[16]
17	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[17]
18	$[-Z, X, Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[18]
19	$[Z, X, -Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[19]
20	$[Z, -X, Y]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[20]
21	$[-Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[21]
22	$[Y, -Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[22]
23	$[-Y, Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[23]
24	$[Y, Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[24]

* Wyckoff site: 12g, site symmetry: 2..

Table 21: Wyckoff bond: 12a@12g, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[1,-4]
2	$[0, -X, Y]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[2,-3]
3	$[Y, 0, X]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[5,-8]
4	$[Y, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[6,-7]
5	$[X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[9,-12]
6	$[-X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[10,-11]
7	$[0, -X, -Y]$	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[13,-16]
8	$[0, X, -Y]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[14,-15]
9	$[-Y, 0, -X]$	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[17,-20]
10	$[-Y, 0, X]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[18,-19]
11	$[-X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[21,-24]
12	$[X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[22,-23]

Table 22: Wyckoff bond: 12b@12g, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[1,4]
2	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[2,3]
3	$[0, X, 0]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[5,8]
4	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[6,7]

continued ...

Table 22

No.	vector	center	mapping
5	$[0,\ 0,\ X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$	[9,12]
6	$[0,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[10,11]
7	$[-X,\ 0,\ 0]$	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[13,16]
8	$[X,\ 0,\ 0]$	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	[14,15]
9	$[0,\ -X,\ 0]$	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[17,20]
10	$[0,\ X,\ 0]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[18,19]
11	$[0,\ 0,\ -X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$	[21,24]
12	$[0,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[22,23]

Table 23: Wyckoff bond: 24c@12g, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[1]
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[2]
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[3]
4	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[4]
5	$[Z,\ X,\ Y]$	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	[5]
6	$[Z,\ -X,\ -Y]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[6]
7	$[-Z,\ -X,\ Y]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[7]
8	$[-Z,\ X,\ -Y]$	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	[8]
9	$[Y,\ Z,\ X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$	[9]
10	$[-Y,\ Z,\ -X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[10]
11	$[Y,\ -Z,\ -X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	[11]
12	$[-Y,\ -Z,\ X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$	[12]
13	$[-X,\ -Y,\ -Z]$	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[13]
14	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	[14]
15	$[X,\ -Y,\ Z]$	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	[15]
16	$[-X,\ Y,\ Z]$	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[16]
17	$[-Z,\ -X,\ -Y]$	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[17]
18	$[-Z,\ X,\ Y]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[18]
19	$[Z,\ X,\ -Y]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[19]
20	$[Z,\ -X,\ Y]$	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[20]
21	$[-Y,\ -Z,\ -X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$	[21]
22	$[Y,\ -Z,\ X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[22]
23	$[-Y,\ Z,\ X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[23]
24	$[Y,\ Z,\ -X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$	[24]

* Wyckoff site: 24h, site symmetry: 1

Table 24: Wyckoff bond: 24a@24h, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	[1]
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[2]
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[3]
4	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[4]

continued ...

Table 24

No.	vector	center	mapping
5	$[Z,\, X,\, Y]$	$[z,\, x,\, y]$	[5]
6	$[Z,\, -X,\, -Y]$	$\left[z,\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[6]
7	$[-Z,\, -X,\, Y]$	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y\right]$	[7]
8	$[-Z,\, X,\, -Y]$	$\left[\frac{1}{2}-z,\, x,\, \frac{1}{2}-y\right]$	[8]
9	$[Y,\, Z,\, X]$	$[y,\, z,\, x]$	[9]
10	$[-Y,\, Z,\, -X]$	$\left[\frac{1}{2}-y,\, z,\, \frac{1}{2}-x\right]$	[10]
11	$[Y,\, -Z,\, -X]$	$\left[y,\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[11]
12	$[-Y,\, -Z,\, X]$	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x\right]$	[12]
13	$[-X,\, -Y,\, -Z]$	$[-x,\, -y,\, -z]$	[13]
14	$[X,\, Y,\, -Z]$	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[14]
15	$[X,\, -Y,\, Z]$	$\left[x+\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	[15]
16	$[-X,\, Y,\, Z]$	$\left[-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[16]
17	$[-Z,\, -X,\, -Y]$	$[-z,\, -x,\, -y]$	[17]
18	$[-Z,\, X,\, Y]$	$\left[-z,\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[18]
19	$[Z,\, X,\, -Y]$	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, -y\right]$	[19]
20	$[Z,\, -X,\, Y]$	$\left[z+\frac{1}{2},\, -x,\, y+\frac{1}{2}\right]$	[20]
21	$[-Y,\, -Z,\, -X]$	$[-y,\, -z,\, -x]$	[21]
22	$[Y,\, -Z,\, X]$	$\left[y+\frac{1}{2},\, -z,\, x+\frac{1}{2}\right]$	[22]
23	$[-Y,\, Z,\, X]$	$\left[-y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[23]
24	$[Y,\, Z,\, -X]$	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, -x\right]$	[24]